

Skill 2 :

1. To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

```
hp@Hima: ~/SKILL
GNU nano 8.7.1
#include <stdio.h>
#include <string.h>

int main() {
    char command[100];

    while (1) {
        // Display prompt
        printf("myshell> ");
        fflush(stdout);

        // Read user input
        if (fgets(command, sizeof(command), stdin) == NULL)
            break;

        // Remove newline
        command[strcspn(command, "\n")] = '\0';

        // Exit condition
        if (strcmp(command, "exit") == 0) {
            printf("Exiting shell...\n");
            break;
        }

        // Ignore empty input
        if (strlen(command) == 0)
            continue;

        // Display entered command
        printf("You entered: %s\n", command);
    }

    return 0;
}
```

```
Child process completed.
hp@Hima:~/SKILL$ nano Skill2.c
hp@Hima:~/SKILL$ gcc Skill2.c -o Skill2
hp@Hima:~/SKILL$ ./Skill2
myshell> hello
You entered: hello
myshell> ls
You entered: ls
myshell> pwd
You entered: pwd
myshell> exit
Exiting shell...
hp@Hima:~/SKILL$ |
```

To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction

```
GNU nano 8.7.1
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];
    int length;

    while (1) {
        printf("myshell> ");
        fflush(stdout);

        if (fgetc(input, sizeof(input), stdin) == NULL)
            break;

        // Process Enter key / remove newline
        input[strcspn(input, "\n")] = '\0';

        // Handle empty input
        if (strlen(input) == 0)
            continue;

        // Exit command
        if (strcmp(input, "exit") == 0) {
            printf("Exiting shell...\n");
            break;
        }

        // Input buffer length
        length = strlen(input);

        printf("Command: %s\n", input);
        printf("Characters entered: %d\n", length);
    }

    return 0;
}
```

```
Exiting shell...
hp@Hima:~/SKILL$ nano Skill2.1.c
hp@Hima:~/SKILL$ gcc Skill2.1c -o Skill2.1
/usr/bin/x86_64-linux-gnu-ld.bfd: cannot find Skill2.1c: No such file or directory
collect2: error: ld returned 1 exit status
hp@Hima:~/SKILL$ gcc Skill2.1.c -o Skill2.1
hp@Hima:~/SKILL$ ./Skill2.1
-bash: ./Skill2.1: No such file or directory
hp@Hima:~/SKILL$ ./Skill2.1
myshell> hello
Command: hello
Characters entered: 5
myshell> hima
Command: hima
Characters entered: 4
myshell> hello.c
Command: hello.c
Characters entered: 7
myshell> |
```

```
hp@Hima: ~/SKILL
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <string.h>

int main()
{
    char command[100];

    while(1)
    {
        printf("2520030141_SHELLFORGE$ ");
        fgets(command, sizeof(command), stdin);

        command[strlen(command) - 1] = '\0';

        if(strcmp(command, "exit") == 0)
            return 0;

        printf("you entered %s\n", command);
    }

    return 0;
}
```

```
hp@Hima:~/SKILL$ nano SimpleCommandloop.c
hp@Hima:~/SKILL$ gcc SimpleCommandloop.c -o SimpleCommandloop
hp@Hima:~/SKILL$ ./SimpleCommandloop
2520030141_SHELLFORGE$
you entered
2520030141_SHELLFORGE$
you entered
2520030141_SHELLFORGE$ exit
hp@Hima:~/SKILL$ |
```